Light

Memory & Storage

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OCR GCSE Computer Science — **Memory and Storage (J277/01 Section 1.2)**. This topic covers primary storage (RAM and ROM), virtual memory, secondary storage types (magnetic, optical, solid state), and how to choose appropriate storage for different scenarios based on characteristics like capacity, speed, portability, durability, and cost. Exam questions typically ask you to **describe** the difference between RAM and ROM, **explain** the purpose of virtual memory, **compare** storage types for given scenarios, and **justify** storage choices for contexts like musicians, embedded systems, or portable devices. Questions range from 1-mark definitions to 4-mark scenario-based justifications.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and apply it to real-world scenarios.

- **Describe** the difference between RAM and ROM, including their volatility and purpose.
- **Explain** the need for primary storage and why computers need both RAM and ROM.
- **Explain** the purpose of virtual memory and when it is used.
- **Describe** the three types of secondary storage: magnetic, optical, and solid state.
- **State** examples of each storage type and their common uses.

- **Compare** storage devices using characteristics: capacity, speed, portability, durability, reliability, and cost.
- **Justify** the choice of storage device for a given scenario (e.g. musician, embedded system, backup).

> **Specification note.** OCR J277 Section 1.2 covers primary storage (RAM, ROM, virtual memory) and secondary storage (types and characteristics). The September 2024 update clarified that questions may ask about storage in the context of embedded systems. Cache is covered in the CPU Architecture topic.

Toolkit

Primary vs Secondary storage — key distinction

Primary storage is directly accessible by the CPU and holds data currently in use. It includes RAM, ROM, and cache. Primary storage is fast but limited in capacity.
Secondary storage is non-volatile storage used to store data permanently when the computer is turned off. It includes hard drives, SSDs, USB sticks, and optical discs. Secondary storage has larger capacity but slower access than primary storage.

RAM vs ROM comparison

Feature	RAM (Random Access Memory)	ROM (Read Only Memory)
Volatility	Volatile — loses contents when power is off	Non-volatile — retains contents without power
Read/Write	Can be read from and written to	Can only be read from (not written to in normal use)
Purpose	Stores currently running programs and data	Stores boot-up instructions / BIOS / firmware
Contents change?	Contents change constantly during use	Contents are permanent (set during manufacture)

Secondary storage types

Type	Examples	How it works
Magnetic	Hard Disk Drive (HDD), magnetic tape	Data stored as magnetic patterns on spinning platters; read/write head moves across surface
Optical	CD, DVD, Blu-ray	Data stored as pits and lands on disc surface; laser reads the pattern of reflections

Type	Examples	How it works
Solid State	SSD, USB flash drive, SD card	Data stored in flash memory chips using electrical charges; no moving parts

Storage characteristics — what to compare

Characteristic	Definition	Exam application
Capacity	How much data can be stored	HDDs typically offer highest capacity for the price
Speed	How fast data can be read/written	SSDs are faster than HDDs (no moving parts)
Portability	How easy to transport	USB drives and SD cards are highly portable
Durability	Resistance to physical damage	SSDs more durable than HDDs (no moving parts to break)
Reliability	How consistently it works over time	SSDs generally more reliable; HDDs can fail mechanically
Cost	Price per unit of storage	HDDs cheaper per GB than SSDs

Exam tip: When comparing storage for a scenario, always link the characteristic to the specific context. Don't just say "SSD is faster" — say "SSD has faster access speed, which means the musician can load audio files more quickly during a live performance."

1. Primary storage: RAM, ROM and virtual memory

Primary storage refers to memory that the CPU can access directly. This includes RAM (for current work), ROM (for start-up instructions), and cache (for frequently used data). Primary storage is essential because the CPU cannot process data directly from secondary storage — data must first be loaded into primary storage.

1.1 RAM (Random Access Memory)

RAM is the computer's main working memory. When you open a program or file, it is loaded from secondary storage into RAM so the CPU can access it quickly. RAM is **volatile**, meaning all data is lost when the computer is turned off.

The amount of RAM affects how many programs can run simultaneously and how smoothly they operate. More RAM allows more data to be held ready for the CPU, reducing the need to fetch data from slower secondary storage.

Why RAM is needed: The CPU works much faster than secondary storage. If the CPU had to fetch data directly from a hard drive every time, the computer would be extremely slow. RAM acts as a fast temporary workspace between the CPU and secondary storage.
How RAM is used: When you open an application, the operating system copies the program code from secondary storage into RAM. As you work, your data (documents, images, etc.) is also held in RAM. When you save your work, data is copied back to secondary storage for permanent keeping.
RAM capacity: Modern computers typically have 8GB to 32GB of RAM. More demanding tasks (video editing, gaming, running multiple applications) benefit from more RAM. If RAM is insufficient, the computer uses virtual memory, which is slower.

1.2 ROM (Read Only Memory)

ROM is **non-volatile** memory that retains its contents even when the power is off. Unlike RAM, ROM cannot be easily written to — its contents are typically set during manufacture and remain permanent.
ROM stores the **BIOS** (Basic Input/Output System) or **UEFI** — the small program that runs when you first turn on the computer. This program performs a **Power-On Self Test (POST)** to check the hardware is working, then locates and loads the operating system from secondary storage into RAM.
The boot process:

1. Power is turned on — CPU starts executing instructions from ROM
2. BIOS/UEFI runs POST to check hardware (RAM, keyboard, drives)
3. BIOS locates the bootable drive (HDD/SSD)
4. Operating system is loaded from secondary storage into RAM
5. Control is handed to the operating system

Why ROM is essential: When you turn on a computer, RAM is empty — there are no instructions for the CPU to follow. ROM provides the initial instructions that get the computer started. Without ROM, the computer couldn't boot.
In embedded systems: ROM (or flash memory programmed like ROM) stores the **firmware** — the permanent software that controls the device. For example, a washing machine's ROM contains the program that controls its cycles. A router's ROM contains its operating software.

1.3 Virtual memory

Virtual memory is a section of secondary storage (usually the hard drive or SSD) that is used as an extension of RAM when RAM becomes full. The operating system moves data that hasn't been used recently from RAM to virtual memory, freeing up RAM for active tasks.
How it works: The operating system divides memory into small blocks called **pages**. When RAM fills up, pages that haven't been accessed recently are moved to a file on the hard drive (the **swap file** or **page file**). If that data is needed again, it's swapped back into RAM, and other unused pages may be moved out.
When virtual memory is used: If you have many programs open and RAM fills up, the operating system starts using virtual memory. Data is swapped between RAM and virtual memory as needed. You might notice the hard drive light flashing frequently when this happens.
The problem — thrashing: Virtual memory is much slower than RAM because it uses secondary storage. Reading from an HDD can be 100,000 times slower than reading

from RAM. If a computer relies heavily on virtual memory, performance drops significantly — the computer spends more time swapping data than doing useful work. This is sometimes called **thrashing**.

Signs of insufficient RAM: Slow performance when switching between programs, hard drive constantly active, programs taking a long time to respond, and the computer becoming sluggish when many programs are open.

Why virtual memory exists: Virtual memory allows computers to run more programs than would fit in RAM alone, preventing crashes when memory runs out. It provides a safety net, though adding more physical RAM is always a better solution for performance.

> **Embedded systems and virtual memory:** The 2024 exam asked why an embedded system (a "Follow Me" car) would not need virtual memory. The answer: embedded systems run a single dedicated program with limited, predictable data requirements, so RAM is sufficient. They also typically lack secondary storage to create virtual memory.

1.4 Memory allocation

When you open a program, the **operating system allocates memory** — it assigns a portion of RAM for that program to use. Each running application gets its own block of memory to store its data and instructions.

The operating system keeps track of which parts of RAM are in use and which are free. When you close a program, its memory is deallocated (freed up) for other programs to use. If too many programs are open and RAM fills up, the OS starts using virtual memory.

Why this matters: Good memory management prevents programs from interfering with each other's data and ensures the computer runs smoothly. If memory isn't managed properly, programs can crash or the whole system can become unstable.

Worked Example 1 – Describing the difference between RAM and ROM

Question: Describe two differences between RAM and ROM. [4 marks]

Model answer:

1. RAM is volatile, meaning it loses its contents when power is turned off **(1)**, whereas ROM is non-volatile and retains its contents without power **(1)**.
2. RAM can be read from and written to **(1)**, whereas ROM can only be read from (not written to in normal use) **(1)**.

> **Exam tip:** For "describe differences" questions, make sure each difference has two parts — what RAM does AND what ROM does. A single statement like "RAM is volatile" only scores half the available marks.

Worked Example 2 – Explaining the purpose of ROM

Question: Explain why a computer needs ROM. [2 marks]

Model answer:
ROM stores the boot-up/start-up instructions (BIOS) **(1)** which are needed to start the computer and load the operating system before RAM has any data in it **(1)**.

Exam tip: The key point is that ROM is needed **before** anything else can run. When you turn on a computer, RAM is empty — ROM provides the initial instructions.

Worked Example 3 — Explaining virtual memory

Question: Explain the purpose of virtual memory. [2 marks]

Model answer:
Virtual memory uses secondary storage as an extension of RAM when RAM is full **(1)**. This allows more programs/data to be used than would fit in RAM alone **(1)**.

Exam tip: Always mention that virtual memory is used "when RAM is full" — this is the trigger condition that examiners look for.

Worked Example 4 — Embedded systems and ROM

Question: Explain why an embedded system in a washing machine stores its program in ROM. [2 marks]

Model answer:
ROM is non-volatile so the program is retained when the power is off **(1)**. The program does not need to change as the washing machine performs the same fixed functions **(1)**.

Exam tip: For embedded systems, emphasise two points: (1) non-volatile so it works immediately when powered on, and (2) the program is fixed/permanent so doesn't need to be rewritten.

2. Secondary storage types

Secondary storage is used to store data permanently — it retains data even when the computer is turned off. There are three main types: magnetic, optical, and solid state. Each has different characteristics that make it suitable for different purposes.

2.1 Magnetic storage

Magnetic storage uses **magnetised patterns** on a surface to represent data. The most common example is the **Hard Disk Drive (HDD)**, which contains spinning metal platters coated with magnetic material. A read/write head moves across the platters to access data.

How HDDs work: Data is stored as tiny magnetic regions on the platter surface. Each region is magnetised in one of two directions, representing binary 0 or 1. The platters spin at high speed (typically 5,400 or 7,200 RPM), and the read/write head floats just above the surface on a cushion of air. To read data, the head detects the magnetic orientation; to write data, it changes the orientation.

Advantages:

- Very high capacity — HDDs can store several terabytes (up to 20TB)
- Low cost per GB — cheapest form of mass storage
- Well-established, reliable technology
- Good for long-term archival storage

Disadvantages:

- Moving parts — the spinning platters and moving head can be damaged by drops, knocks, or vibration
- Slower than solid state — mechanical movement takes time; the head must physically move to the right location
- Generates heat and uses more power than SSDs
- Noise from spinning platters and moving heads
- Larger physical size than SSDs

Common uses: Desktop computers, servers, data centres, backup storage, storing large media libraries, network attached storage (NAS).

2.2 Optical storage

Optical storage uses a **laser** to read data stored as patterns of pits and lands (flat areas) on a reflective disc. Examples include CD, DVD, and Blu-ray discs.

How optical discs work: Data is stored as microscopic pits burned into the disc surface. A laser beam is shone onto the disc, and the pattern of reflections (pits scatter light differently from flat lands) is detected and converted back into binary data. Writable discs (CD-R, DVD-R) use a dye layer that can be permanently changed by a higher-powered laser.

Types of optical disc:

- **CD** (Compact Disc) — 700 MB capacity; used for music and small data transfers
- **DVD** (Digital Versatile Disc) — 4.7 GB single layer, 8.5 GB dual layer; used for films and software
- **Blu-ray** — 25-50 GB; used for HD/4K video and games; uses a blue laser for higher density

Advantages:

- Very cheap to produce in quantity — good for distributing software, films, and music
- Portable and lightweight
- Not affected by magnetic fields (unlike HDDs)
- Good for archival storage if kept properly

Disadvantages:

- Limited capacity compared to HDD/SSD
- Surface can be scratched, affecting readability

- Slower read/write speeds than HDD or SSD
- Requires an optical disc drive (many modern laptops and tablets don't have one)
- Becoming less common as streaming and digital downloads replace physical media

Common uses: Distributing films, music, games and software; backup archives; sharing data when internet isn't available.

2.3 Solid state storage

Solid state storage uses **flash memory chips** to store data as electrical charges in memory cells. There are no moving parts — hence "solid state". Examples include **SSDs** (Solid State Drives), **USB flash drives**, and **SD cards**.

How SSDs work: Data is stored in memory cells made of transistors. Each cell traps electrons to represent binary values. Because there are no moving parts, data can be accessed almost instantly — the controller simply reads the electrical state of the appropriate cells. This is why SSDs are so much faster than HDDs.

Advantages:

- Very fast read/write speeds — no mechanical delays; data access is almost instant
- Highly durable — no moving parts means they can survive drops, bumps, and vibration
- Silent operation — no spinning discs or moving heads
- Low power consumption — excellent for laptops and portable devices; extends battery life
- Compact and lightweight — can be made very small
- Faster boot times — computers with SSDs start up in seconds rather than minutes

Disadvantages:

- Higher cost per GB than magnetic storage (though prices continue to fall)
- Limited write cycles — flash memory cells wear out after many write operations (though modern SSDs last many years in normal use)
- Lower maximum capacity than HDDs (though this gap is closing rapidly)
- Data recovery can be more difficult if the drive fails

Common uses: Laptops, smartphones, tablets, portable storage (USB drives), cameras (SD cards), gaming consoles, modern desktop computers as the primary drive.

2.4 Capacity comparison

Storage type	Typical capacity
CD	700 MB
DVD	4.7 GB (single layer)
Blu-ray	25—50 GB
USB flash drive	8 GB — 1 TB
SD card	32 GB — 1 TB
SSD	256 GB — 8 TB
HDD	500 GB — 20 TB

Exam tip: Know at least one example of each storage type. The 2024 exam asked candidates to name "a different type of secondary storage" after mentioning magnetic and solid state — optical (CD/DVD/Blu-ray) was the expected answer.

2.5 Cloud storage

Cloud storage stores data on remote servers accessed via the internet. Examples include Google Drive, Dropbox, iCloud, and OneDrive. While not a "type" of storage in the same sense as magnetic/optical/solid state (cloud servers actually use HDDs and SSDs), it's an increasingly common storage solution.

Advantages: Accessible from anywhere with internet; automatic backup; easy sharing; scales easily; no physical hardware to maintain.

Disadvantages: Requires internet connection; ongoing subscription costs; privacy/security concerns; slower than local storage; dependent on provider.

Worked Example 5 — Describing how magnetic storage works

Question: Describe how data is stored on a hard disk drive. [2 marks]

Model answer:
Data is stored as magnetic patterns/areas on spinning platters **(1)**. A read/write head moves across the platters to read or write data **(1)**.

Exam tip: Mention both the magnetic patterns AND the moving parts (spinning platters, read/write head). This shows you understand the mechanical nature of HDDs.

Worked Example 6 — Comparing storage types

Question: Give one advantage of solid state storage over magnetic storage. [1 mark]

Model answer:
Solid state storage has faster access/read/write speed **(1)**.
Alternative answers: More durable (no moving parts to break) / Uses less power / Silent / More portable / Lighter

Exam tip: Speed and durability are the safest answers for SSD advantages. Both relate directly to "no moving parts" — the fundamental difference from HDDs.

Worked Example 7 — Explaining optical storage

Question: Describe how data is read from an optical disc. [2 marks]

Model answer:
A laser is shone onto the disc surface **(1)**. Data is read from the pattern of reflections caused by pits and lands on the disc **(1)**.

Exam tip: Key terms for optical: laser, pits, lands, reflection. Don't confuse with magnetic (which uses magnetism, not light).

Worked Example 8 — Identifying storage types

Question: State whether each of the following is magnetic, optical, or solid state storage: (a) DVD, (b) USB flash drive, (c) Hard disk drive. [3 marks]

Model answer:
(a) DVD — Optical **(1)**
(b) USB flash drive — Solid state **(1)**
(c) Hard disk drive — Magnetic **(1)**

Exam tip: Learn the category for each common device. USB drives and SD cards are solid state (not "removable" or "flash" as a category — flash IS solid state).

3. Choosing storage for different scenarios

Exam questions often present a scenario and ask you to justify which storage type is most suitable. To answer well, you need to match the **characteristics** of each storage type to the **requirements** of the scenario.

3.1 The six characteristics

When comparing storage devices, consider these six characteristics:

Capacity: How much data can be stored. Large media files need high capacity. HDDs offer the most storage for the price.

Speed: How quickly data can be read and written. Fast access is important for loading programs, gaming, and working with large files. SSDs are fastest.

Portability: How easy it is to carry. USB drives and SD cards are highly portable. External HDDs are portable but bulkier.

Durability: Resistance to physical damage. SSDs have no moving parts, so they survive drops better than HDDs.

Reliability: How consistently the device works over time. SSDs have no mechanical parts to wear out, though flash memory has limited write cycles.

Cost: Price per GB of storage. HDDs are cheapest per GB. SSDs cost more but prices are falling.

3.2 Common exam scenarios

Musician storing audio files: Needs high capacity (audio files are large), may need fast access for live playback, may need portability and durability if travelling. SSD or high-capacity HDD both valid choices with different trade-offs.
Backing up important data: Needs high capacity and reliability. Cost may be important for large backups. HDD or cloud storage often suitable.
Laptop for everyday use: Benefits from SSD for speed (fast boot, quick application loading), durability (survives being carried around), and low power consumption (better battery life).
Embedded system: Uses ROM or flash memory for firmware. Needs to be non-volatile, reliable, and compact. Data requirements are usually small and fixed.
Distributing software: Optical discs (DVDs, Blu-rays) are cheap to produce in quantity and can be distributed physically.

> **Exam tip:** The 2024 exam asked candidates to justify choosing solid state OR magnetic storage for a musician. Both answers could gain full marks if properly justified. The key is linking characteristics to the musician's specific needs.

3.3 Understanding storage units

When comparing storage capacity, you need to understand the units and be able to convert between them:

Unit	Size	Conversion
1 Kilobyte (KB)	1,000 bytes	÷ 1,000 to get MB
1 Megabyte (MB)	1,000 KB	÷ 1,000 to get GB
1 Gigabyte (GB)	1,000 MB	÷ 1,000 to get TB
1 Terabyte (TB)	1,000 GB	× 1,000 to get GB

Exam application: The 2024 exam asked "Which is the smallest capacity?" with options including 200,000 KB and 300 MB. To compare, convert to the same unit: 200,000 KB = 200 MB, which is smaller than 300 MB.

3.4 Making justified recommendations

In extended questions, you must **make a clear recommendation** and **justify it with specific reasons**. The examiner's report notes that candidates who say "both are suitable" without making a final choice lose marks.
Structure your answer:

- State advantages of option A for this scenario
- State advantages of option B for this scenario
- Acknowledge any limitations and whether they matter here
- Make a clear recommendation: "I would recommend X because..."
- Summarise your key reasons

> Worked Example 9 — Justifying storage choice (SSD)
> **Question:** A musician needs to store large audio files and wants to take the storage device to live performances. Justify why solid state storage would be suitable. [3

marks]

Model answer:
Solid state has faster access speed, allowing audio files to be loaded quickly during performances **(1)**.
Solid state is more durable with no moving parts, so it can survive being transported without risk of damage **(1)**.
Solid state is lightweight and portable, making it easy to carry to different venues **(1)**.

Exam tip: Give three distinct points, each linking a characteristic (speed, durability, portability) to the scenario (musician, live performances, transport).

Worked Example 10 – Justifying storage choice (HDD)

Question: The same musician could also choose magnetic storage. Justify why this would be suitable. [3 marks]

Model answer:
Magnetic storage (HDD) has higher capacity, allowing more audio files to be stored **(1)**.
Magnetic storage is cheaper per GB, so large storage is more affordable **(1)**.
Although access speed is slower than SSD, it is still fast enough for audio playback **(1)**.

Exam tip: When justifying the "weaker" option, acknowledge its limitations but explain why they don't matter for this scenario. This shows evaluative thinking.

Worked Example 11 – Storage for backup

Question: A business needs to back up 10TB of data. Explain why magnetic storage would be more suitable than solid state for this purpose. [2 marks]

Model answer:
Magnetic storage is cheaper per GB, so backing up 10TB would cost significantly less **(1)**.
Magnetic storage is available in very high capacities (up to 20TB per drive), making it practical for large backups **(1)**.

Exam tip: For backup scenarios, cost and capacity are usually the key factors. Speed is less important for backup than for everyday use.

Worked Example 12 — Extended comparison (4-mark style)

Question: A school is buying new computers for a classroom. Compare magnetic and solid state storage for this use, and recommend which would be more suitable. [4 marks]

Model answer:
Solid state storage (SSD) has faster read/write speeds, so computers would boot up faster and applications would load more quickly, reducing waiting time for students **(1)**.
SSDs are more durable with no moving parts, which is important in a school environment where computers may be knocked or bumped **(1)**.
Magnetic storage (HDD) is cheaper per GB, which could allow the school to buy more computers within their budget, or provide more storage per computer **(1)**.
I would recommend SSDs because the faster performance and durability outweigh the higher cost, and school computers typically don't need very large storage capacity **(1)**.

Exam tip: For comparison questions, discuss BOTH options with pros/cons, then give a clear recommendation with justification. Don't sit on the fence — make a decision and explain why.

Worked examples summary

The 12 worked examples above cover the main question types you will encounter on memory and storage. Here is a summary of the key question patterns:

Example	Question type	Key skill
1	Describe differences (4 marks)	RAM vs ROM — volatility and read/write
2	Explain purpose (2 marks)	ROM stores boot-up instructions
3	Explain purpose (2 marks)	Virtual memory extends RAM
4	Explain with context (2 marks)	Embedded systems use ROM
5	Describe how it works (2 marks)	Magnetic storage mechanism
6	Give advantage (1 mark)	SSD vs HDD comparison

Example	Question type	Key skill
7	Describe how it works (2 marks)	Optical storage mechanism
8	Identify types (3 marks)	Classify storage devices
9	Justify choice (3 marks)	SSD for musician scenario
10	Justify choice (3 marks)	HDD for musician scenario
11	Explain suitability (2 marks)	HDD for backup scenario
12	Compare and recommend (4 marks)	School scenario — evaluate both options

Quick reference

Key terms – exam-ready definitions

Term	Definition (mark scheme style)
Primary storage	Memory directly accessible by the CPU; holds currently used data and instructions (RAM, ROM, cache)
Secondary storage	Non-volatile storage for permanent data; not directly accessible by CPU (HDD, SSD, optical)
RAM	Volatile memory that stores currently running programs and data; contents lost when power is off
ROM	Non-volatile memory that stores permanent instructions (BIOS/firmware); cannot be written to in normal use
Virtual memory	Section of secondary storage used as an extension of RAM when RAM is full
Volatile	Memory that loses its contents when power is turned off
Non-volatile	Memory/storage that retains contents without power
Magnetic storage	Storage using magnetic patterns on spinning platters (e.g. HDD)
Optical storage	Storage using laser to read pits/lands on a reflective disc (e.g. CD, DVD, Blu-ray)
Solid state storage	Storage using flash memory chips with no moving parts (e.g. SSD, USB drive, SD card)
BIOS	Basic Input/Output System; boot-up instructions stored in ROM

Term	Definition (mark scheme style)
Firmware	Permanent software stored in ROM that controls an embedded system
Cloud storage	Data stored on remote servers accessed via the internet (e.g. Google Drive, Dropbox)
Memory allocation	The process by which the OS assigns portions of RAM to running programs

Storage comparison — quick reference

Characteristic	Magnetic (HDD)	Solid State (SSD)	Optical (DVD)
Capacity	Very high (up to 20TB)	High (up to 8TB)	Low (4.7GB—50GB)
Speed	Medium	Very fast	Slow
Durability	Low (moving parts)	High (no moving parts)	Medium (can scratch)
Cost per GB	Low (cheapest)	Medium-high	Very low for discs
Portability	Medium (bulky)	High (compact)	High (thin, light)
Power use	Higher	Lower	Medium

RAM vs ROM — quick comparison

Feature	RAM	ROM
Volatile?	Yes — loses data when power off	No — keeps data without power
Can write?	Yes — read and write	No — read only
Contents	Currently running programs/data	Boot-up instructions / BIOS
Changes?	Constantly during use	Permanent (set at manufacture)

Typical pitfalls

Pitfall 1 — Saying RAM is "permanent" or ROM is "temporary".
Wrong: "ROM is temporary memory."
Right: ROM is **non-volatile** (permanent). RAM is **volatile** (temporary — loses data when power is off).

Pitfall 2 — Thinking ROM can be written to.
Wrong: "The computer saves files to ROM."
Right: ROM is **read-only** — you cannot write data to it in normal use. Files are saved to secondary storage.

Pitfall 3 — Confusing virtual memory with secondary storage.
Wrong: "Virtual memory is the hard drive."
Right: Virtual memory is a **section of** secondary storage used to extend RAM — not the entire hard drive.

Pitfall 4 — Not explaining WHY virtual memory is slower.
Wrong: "Virtual memory is slow."
Right: Virtual memory is slow **because it uses secondary storage**, which has much slower access speed than RAM.

Pitfall 5 — Thinking SSD has unlimited lifespan.
Wrong: "SSDs last forever and never fail."
Right: SSDs have limited write cycles (flash memory cells wear out), though modern SSDs last many years in normal use.

Pitfall 6 — Not linking characteristics to the scenario.
Wrong: "SSD is faster so it's better."
Right: "SSD is faster, **which means the musician can load audio files quickly during live performances**."

Pitfall 7 — Forgetting optical as a storage type.
Wrong: "The two types of secondary storage are magnetic and solid state."
Right: There are **three** types: magnetic, solid state, and **optical** (CD, DVD, Blu-ray).

Pitfall 8 — Calling USB drives "memory sticks" without clarifying type.
Wrong: "A memory stick is removable storage."
Right: A USB flash drive is **solid state** storage — it uses flash memory chips with no moving parts.

Pitfall 9 — Thinking more RAM eliminates the need for virtual memory.
Wrong: "If you have 32GB RAM you never need virtual memory."
Right: Virtual memory is a safety net when RAM fills up. More RAM reduces reliance on it, but the OS may still use it.

Pitfall 10 — Incomplete RAM/ROM comparisons.
Wrong: "RAM is volatile" (only describes RAM).

Right: "RAM is volatile **whereas ROM is non-volatile**" (describes both sides of the comparison).

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